AI PLAYBOOK for PUBLIC HEALTH

AI Playbook and Toolkit for Public Health Departments

MLOps and LLMOps for Public Health

OPS 300

MLOps and LLMOps are the lifecycle practices used to develop, test, deploy, monitor, update, and retire AI systems. Public health agencies need these practices because AI performance can change over time as data, reporting patterns, policies, users, and workflows change. This module focuses on model registries, versioning, testing, drift monitoring, evaluation datasets, release controls, rollback, and post-deployment governance.

Level	intermediate/practitioner
Primary track	Operations and Data Quality Track
Appears in tracks	operations-data-quality

Learning Objectives

- Explain the purpose of mlops and llmops for public health in public health AI work.
- Identify the technical, operational, privacy, security, equity, and governance issues associated with the topic.
- Apply the topic to a real or realistic public health workflow, system, or implementation scenario.
- Recognize common failure modes that can affect safety, accuracy, trust, workflow fit, or sustainability.
- Identify practical guardrails that should be documented before implementation.
- Produce a AI operations plan that can support readiness assessment, governance review, procurement, implementation, or monitoring.

Module Overview

MLOps and LLMOps are the lifecycle practices used to develop, test, deploy, monitor, update, and retire AI systems. Public health agencies need these practices because AI performance can change over time as data, reporting patterns, policies, users, and workflows change. This module focuses on model registries, versioning, testing, drift monitoring, evaluation datasets, release controls, rollback, and post-deployment governance.

Why This Topic Matters for Public Health AI

MLOps and LLMOps for Public Health matters because AI systems are embedded in public health programs, data systems, and decision processes rather than operating in isolation. The topic affects whether AI outputs are accurate, explainable, safe, equitable, secure, and sustainable. Agencies that ignore this area may create tools that work in a demonstration but fail in actual practice.

The topic also affects accountability. Public health agencies must be able to explain how data are used, why a tool was approved, what evidence supports the workflow, who reviews outputs, and what happens when the system fails. These responsibilities become more important as AI tools move from experimentation into operational use.

Core Concepts

MLOps refers to operational practices for managing machine learning models through their lifecycle. LLMOps applies similar practices to large language model applications, including prompts, retrieval sources, evaluation datasets, guardrails, and model versions. These practices make AI systems more reliable, auditable, and maintainable.

Model versioning is the practice of tracking which version of a model, prompt, retrieval corpus, evaluation dataset, or configuration was used. Versioning matters because public health staff may need to reconstruct why an output changed or why an incident occurred.

Drift occurs when data patterns, user behavior, or model performance change over time. In public health, drift may occur because a new lab begins reporting, a case definition changes, a provider changes coding practices, or community conditions shift. Monitoring drift helps agencies decide when retesting or retraining is needed.

Public Health Example

A syndromic surveillance team may use an anomaly detection model to flag unusual emergency department visit patterns. The model may perform well during testing, but performance can change when chief complaint text changes, new facilities join the feed, or seasonal patterns shift. An MLOps plan would define monitoring indicators, evaluation frequency, responsible staff, retraining rules, and rollback procedures.

Technical Deep Dive

Operational AI management starts before deployment. Teams should define an intended use, baseline performance, evaluation datasets, release criteria, monitoring indicators, and an owner responsible for review. Without these elements, agencies may not know whether the deployed system is still behaving as approved.

LLMOps adds additional elements because generative systems depend on prompts, retrieval documents, context windows, model settings, safety filters, and human review workflows. A change in any of these components can change outputs. Agencies should track prompt templates, source document updates, model upgrades, and evaluation results.

Rollback is a critical safeguard. If a new model version, prompt, data pipeline, or retrieval index creates errors, staff need a documented way to return to the prior approved version or disable the AI function. Rollback authority should be defined before deployment.

Risks, Failure Modes, and Guardrails

A common failure mode is treating the topic as a technical detail rather than a public health control. When staff do not connect technical decisions to authority, workflow, equity, privacy, security, and accountability, the agency may adopt a tool that appears useful but cannot be sustained or defended.

Another risk is relying on vendor claims without agency-defined evidence. Vendors may provide demonstrations, dashboards, or summary metrics that do not match the agency workflow or data environment. A guardrail is to require documentation, test results, acceptance criteria, and agency-controlled review.

A third risk is failing to plan for change. Public health data, policies, systems, and emergencies change. Any technical approach must include monitoring, review, versioning, escalation, and retirement pathways so the agency can respond when assumptions no longer hold.

Application to AI-Supported Workflows

Generative AI, predictive analytics, RAG, automation, and agentic workflows all depend on the controls described in this module. The controls should be scaled to the risk of the use case. A tool that drafts internal training text may require lighter review than a tool that updates case records, prioritizes outreach, or supports public communication during an emergency.

The module should be applied during readiness assessment, procurement, implementation planning, pilot testing, and post-deployment monitoring. Learners should document how the topic affects data, systems, users, safeguards, and decision-making for the selected workflow.

Reflection Questions

Where does this topic appear in one current or proposed AI workflow in your agency?

What decisions, data sources, users, or partners would be affected if this area is poorly designed?

What evidence would governance reviewers need before approving the workflow?

Which safeguards should be required before pilot testing?

Who owns ongoing monitoring and review?

Practical Exercise

Create a AI operations plan for one real or realistic public health AI workflow. The artifact should describe the use case, affected users, data sources, system dependencies, risks, guardrails, review points, and unresolved questions.

The exercise should produce a work product that could be used in a governance meeting, procurement review, implementation planning session, pilot evaluation, or monitoring review. Learners should document assumptions and identify which staff or partners need to be consulted.

Expected Artifact or Evidence

The expected artifact is a completed AI operations plan. It should be specific to a public health workflow and should include technical dependencies, governance questions, human review expectations, monitoring indicators, and unresolved risks.

Expected Assignment

- The expected artifact is a completed AI operations plan. It should be specific to a public health workflow and should include technical dependencies, governance questions, human review expectations, monitoring indicators, and unresolved risks.

Knowledge Check

- 1. What is the best reason to address mlops and llmops for public health before deployment?
- 2. Which practice best supports responsible implementation?
- 3. Which statement best describes a common failure mode?
- 4. What should be included in the practical artifact for this module?
- 5. When should the issue be escalated for governance or leadership review?

References and Resources

- NIST AI Risk Management Framework: https://www.nist.gov/itl/ai-risk-management-framework
- Platform-specific MLOps and LLMOps documentation for approved agency tools
- NIST Generative AI Profile: https://www.nist.gov/publications/artificial-intelligence-risk-management-framework-generative-artificial-intelligence
- CDC Public Health Data Strategy and Data Modernization resources: https://www.cdc.gov/public-health-data-strategy/php/about/index.html
- OWASP Top 10 for LLM Applications, when security or AI integrations are relevant
- Agency privacy, cybersecurity, data governance, procurement, and records management policies